NILS JÖRGENSEN

Autonomous Systems Researcher | Motion & Mission Planning | Industrial Robotics

Sofielundsvägen 5, 121 32 Enskededalen, Sweden   070-581 97 06   @ nilsjor@kth.se

EXPERIENCE

Ph. D. Researcher | Networked Industrial Robotics
KTH Royal Institute of Technology
Autumn 2020 – Spring 2026   Stockholm, Sweden

Research spanning task-, motion-, and mission planning for safe industrial multi-robot coordination, including PDDL, AI-planning, and MILP optimization. Surveyed planning algorithms for indoor robotics and developed joint planners integrating task scheduling with wireless network resource allocation, focusing on logistics.

Teacher, Student Supervisor, and Research Engineer
KTH | Department of Machine Design
Autumn 2019 – Spring 2026   Stockholm, Sweden

Years of teaching, course development, and research, covering computer architectures, assembly and C/C++ bare-metal and RTOS programming, distributed systems, modeling and control of dynamical systems, and everything in between. Supervised countless in degree projects as well as industrial proof-of-concept projects within the Mechatronics capstone course.

EDUCATION

Ph. D. in Mechatronics and Embedded Control Systems
KTH Royal Institute of Technology
Autumn 2020 – Spring 2026   Stockholm, Sweden

Formal Methods & Software Verification • Hybrid Systems • Networked Control Systems • Cyber-Physical Security • Reinforcement Learning • Convex Optimization • Fault-Tolerant Design • Research Methodology • Scientific Writing

M. Sc. in Engineering Design | Mechatronics Track
KTH Royal Institute of Technology
Autumn 2017 – Spring 2019   Stockholm, Sweden

Internetworking • Internet Security and Privacy • Embedded and Distributed Control Systems • Nonlinear Control • Optimal and Model-Predictive Control • Low-Level Programming • Robust Electronics • Test-Driven Development • Systems Engineering

B. Sc. in Mechanical Engineering | International Program
RWTH Aachen University / KTH Royal Institute of Technology
Autumn 2014 – Summer 2017   Aachen, Germany

Automatic Control • Object-Oriented Programming • Algorithms and Optimisation • Statistics and Probability Theory • Numerical Analysis • Machine Components • Electrical Engineering • Physics

LANGUAGES

Swedish
English
German

TECHNICAL SKILLS

Planning & optimization
Mission & motion planning with a focus on PDDL, MILP, and other optimization-based methods.

Networked multi-robot systems
Multi-robot task allocation, and QoS-aware replanning in dynamic environments. Joint optimization of task and resource scheduling.

Formal methods & systems theory
Model checking, satisfiability solving, hybrid-systems analysis. Nonlinear and optimal control.

C++, Python, MATLAB and ROS2
Modeling, simulation, prototyping and deployment: bare-metal, RTOS, and application-level development.

PUBLICATIONS

Joint 5G-aware task planner for industrial multi-robot use-case
10.1109/ETFA54631.2023.10275420

Survey of communication-aware robot planning algorithms for 5G
10.1109/ETFA52439.2022.9921449

Robotic measurement campaign & Gaussian process model validation
10.48550/arXiv.2603.08865

Trustworthy edge computing for cyber-physical systems: a review
10.1145/3539662

REFERENCES

Available upon request.